EtherCAT Interface Software: Quick Start Guide

1. Overview

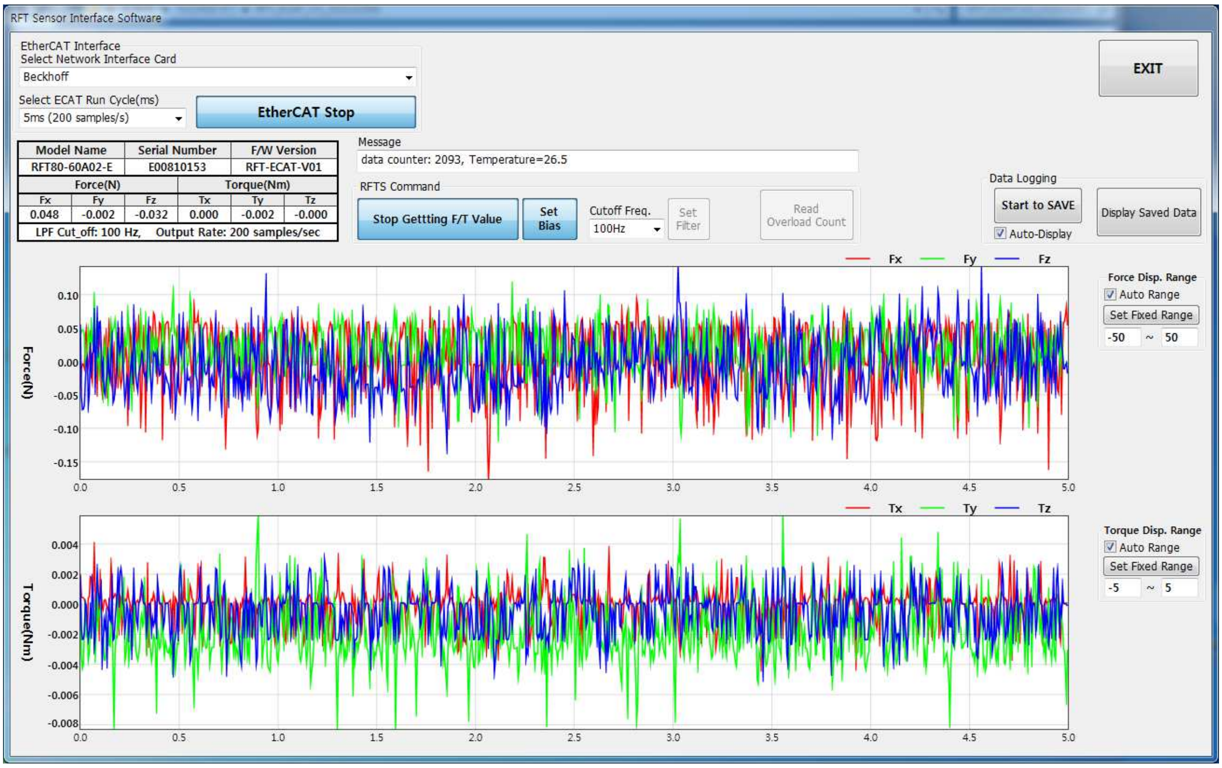
The RFT Series Sensor Interface Software is designed to monitor and control force/torque sensors via EtherCAT communication. It provides real-time data visualization, sensor configuration, and logging capabilities.

2. Prerequisites

To ensure the RFT Series Sensor Interface Software functions correctly, the following network library must be installed on your Windows-based PC:

1) The software requires either **WinPcap** or **Npcap** to capture EtherCAT frames directly through the local network stack for high-speed data acquisition.
2) If you choose to install Npcap, you must enable the "**Install Npcap in WinPcap API-compatible Mode**" option during the installation process.
3) Do not install both WinPcap and Npcap simultaneously. Please completely uninstall any existing versions before installing a new driver.

3. Main Interface Description



4) EtherCAT Interface Setup
   * Select Network Interface Card
   * Choose the NIC connected to the EtherCAT network.
   * Select **ECAT Run Cycle** (ms). [Example: 5 ms (200 samples/sec)]
   * **Start ECAT** to Initialize EtherCAT communication with the sensor.

5) Sensor Information Panel
   Once communication is established, this information is automatically updated.

   * Model Name | Serial Number| Firmware Version

6) Real-Time Data Display
   Once data acquisition starts, the measured values are automatically updated in real time.

   * Real-time force/torque data: Fx, Fy, Fz, Tx, Ty, Tz
   * Upper graph: Force (N)
   * Lower graph: Torque (Nm)

7) RFTS Command Panel
The RFTS Command Panel provides control and configuration functions for the sensor.

   * **Get F/T Value Continuously**:
     Starts/stops real-time streaming of force/torque data (Toggle)
   * **Set Bias**:
     Toggles zero setting (offset removal) on/off
   * **Cut-off Frequency**:
     Selects LPF cut-off frequency [e.g., 1Hz ~ 1kHz(no filter)]
   * **Set Filter**:
     Applies the selected cutoff frequency
   * **Read Overload Count**:
     Displays the number of overload events detected by the sensor

8) Data Logging
   * **Start to SAVE** to begin recording sensor data (Toggle)
   * Auto-Display (✓): automatically shows logged data after logging
   * Display Saved Data: opens previously recorded data for review

9) Display Range Settings
   * Force / Torque Display Range: Adjustable (Auto/Manual)
   * Auto Range (✓): automatically scales to fit input data
   * Set Fixed Range: Displays data within the set range only

10) Exit
   * Click EXIT to safely close the application

4. Basic Operation Procedure

❶ Connect the sensor to the PC via EtherCAT

❷ Launch the software

❸ Select the correct Network Interface Card

❹ Set the desired ECAT Run Cycle

❺ Click Start ECAT

❻ Verify that sensor information is displayed

❼ Click Get F/T Value Continuously to start measurement

❽ (Optional) Click Set Bias to zero the sensor

❾ Adjust filter and display settings as needed

❿ Use Start to SAVE for data logging